

PROBLEM STATEMENT



KPI's REQUIREMENT

- **YTD Sales:** Monitor year-to-date sales to gauge the overall revenue performance over time.
- **QTD Sales:** Track quarterly sales figures to identify sales trends and fluctuations.
- **YTD Products Sold:** Analyse the total number of products sold throughout the year to understand product movement.
- **YTD Reviews:** Keep tabs on year-to-date product reviews to assess customer feedback and satisfaction.

- **Sales by Month (Line Chart):** Visualize sales trends over time on a monthly basis to identify seasonal patterns and growth trends.
- **Sales by Week (Column Chart):** Display sales data on a weekly basis to pinpoint shorter-term fluctuations and performance insights.
- **Sales by Product Category (Text/Heat Map):** Utilize a text or heat map visualization to provide a high-level overview of sales across different product categories.
- **Top 5 Products by YTD Sales (Bar Chart):** Highlight the top-performing products based on year-to-date sales to focus on key revenue generators.
- **Top 5 Products by YTD Reviews (Bar Chart):** Identify the top-rated products by year-to-date reviews to understand customer preferences.

1. Check Data Quality (Data Transformation)
2. Data Cleaning(null values, Date colm)

Make Date Table

Whenever a client has a requirement of YTD or other Date related visualization, recommended to make and use a Date Table (Design Decision) → As it makes the use of the Time Intelligence Functions quite easy and handy to use.

```
1 Date Table = CALENDAR(MIN(Amazon_Data[Order Date]), MAX(Amazon_Data[Order Date]))
```

(Makes the date column with start date (MIN) and end date (MAX) of the available order dates)

Extract YQMD(Year, Quarter, Month, Day)

From the extracted data of the date, we have to extract other Exploratory features like Year, Quarter, Month, Day to gain deep insights on each DAY, MONTH, QUARTER, for the YTD and the QTD sales

```
1 Month = FORMAT('Date Table'[Date], "MMM")
```

Create Relationship between the two tables to flow data from one table to another on the basis of the common columns in both

→ Creating data model and relation between the two tables on **DATE** as it is common in both so that, we can use and model the data on both the tables using the data from both the tables.

(Explain the concept of PK and FK here between the two tables)

i. 1→many relation ship is recommended

ii. Many→many relationship is not recommended as the data size will excessively increase in this case

VISUALIZATION

- 1) Write DAX on the main table(fact table) using Time Intelligence function to get the YTD and QTD sales using the Date table → Create new **Measure** for the KPI

```
YTD Sales = TOTALYTD(SUM(Amazon_Data[Price(Dollar)]), 'Date Table'[Date])
```

TOTALYTD(Expression for which YTD is to be calculated, For which date it is to be calculated)

Similarly:

```
QTD Sales = TOTALQTD(SUM(Amazon_Data[Price(Dollar)]), 'Date Table'[Date])
```

YTD Products Sold : We need to find the number of products sold in YTD

```
1 YTD Products Sold = TOTALYTD(COUNT(Amazon_Data[Product Category]), 'Date Table'[Date])
```

COUNT → Doesnot count the blank values/sales

So we use the Product Category here as an expression to count the products sold as during the Data Quality Check, we saw that Product Category has 100% valid fields, so it will work the best with COUNT as it doesnt consider blank fields.

CHARTS

- 1) **SALES BY MONTH:** Using area chart to show Sales by Month (X→ Month, Y→ Sales).

Sorting by month: Add a Month number column in the Date Table to sort by month (Jan → Dec)

We need this as we need to sort the area chart by Month

```
1 Month Number = MONTH('Date Table'[Date])
```

(This will assign a specific month number to each month by using the Date from the Date table)

After creating this, go to report view, select the Month measure under the Date Table > Column Ribbon > Sort Column by Month Number

Select the chart, sort by : Month, Ascending (Jan→Dec)

- 2) **SALES BY WEEK:** Similarly, we will need a week number that we need to add to the data table.

Week Number = WEEKNUM('Date Table'[Date])		
nth	Month Number	Week Number
	1	1

- 3) **SALES BY PRODUCT CATEGORY:** We are using a Grid visual to show this, along with conditional formatting to show the highest and lowest values based on Gradient color coding.

ROWS → Product Categories

COLS → YTD Sales, QTD Sales, %YTD Sales

Basically, we are showcasing the top performing categories of products in this

- 4) **TOP 5 PRODUCTS BY YTD SALES:** We need to show the top 5 products by YTD sales. We use a Bar Chart for this

X axis → Product Description

Y axis → YTD Sales

This will show all the products. But we need to show only the top 5. So we use

FILTER pane.

Product Descripti...

top 5 by YTD Sales

Filter type

Top N

Show items

Top 5

By value

YTD Sales

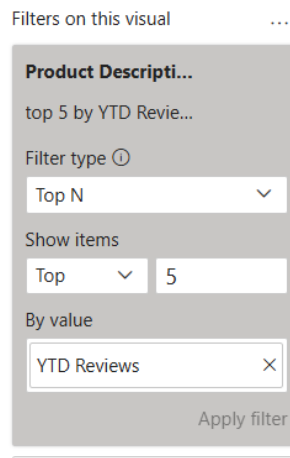
Apply filter

(Use the FILTER BY VISUAL option). Using conditional formatting to get the highest and lowest heatmapped values

- 5) **TOP 5 PRODUCTS BY YTD REVIEWS:** This will be nearly the same as before. Here we use the following:

X axis → Product Description

Y axis → YTD Reviews



FILTERS

To make the dashboard dynamic and responsive to different values, we use slicer(Dropdown) to add filters and find/sort data based on it

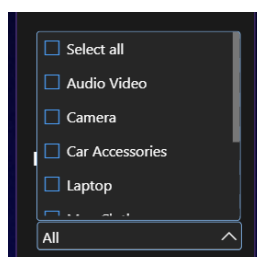
- 1) Filter By **Product Category**: We choose different product categories/items to filter. We use Field→Product Category data
- 2) Filter by **Quarter**: Used to find sales in different quarters. For this we need to create a separate quarter column in the data table to get the quarters
 - Make a Quarter Number column and get the quarters from the date. Use the DAX function as follows:

```
Quarter Number = QUARTER('Date Table'[Date])
```

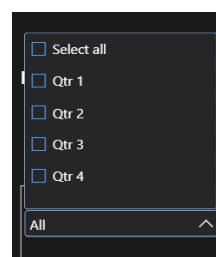
- Make a Quarter Column (that will be used in actual slicer) to show QTR(number) for visual ease. We use the **CONCATENATE** DAX query:

```
Quarter = CONCATENATE("Qtr ", 'Date Table'[Quarter Number])
```

The filters will be something like:



Filter By **Product Cat**



Filter By **QTR**